

Exercise Risk Matrix:

1. Ventilation Rate Impact: Strenuous exercise increases inhalation of ambient pollutants by 4x to 8x.
2. AQI ≤ 50 : All outdoor physical activities, running, and competitive sports safe.
3. AQI 51-100: Safe for healthy athletes. Monitor athletes with asthma or exercise-induced bronchospasm.
4. AQI 101-150: Reduce duration and intensity of outdoor workouts. Substitute high-intensity cardio with indoor activities.
5. AQI > 150 : Cancel or move all outdoor endurance sports and school athletic practices indoors.
6. Timing Optimization: Avoid training near major highways during 07:00-09:00 and 17:00-19:00 peak traffic hours.